ERC Consolidator Grant 2027: Research Proposal, Part B1

Acronym: HOVER

Title: Hoverfly-Optimised Vision through Event-driven Recurrent spiking circuits: from a sexually dimorphic eye to active machine vision

Principal Investigator: Prof. Dr. Étienne Serbe-Kamp

Host institution: IU Internationale Hochschule GmbH, Germany *[confirm exact legal name]*

Proposal duration: 60 months

Primary panel: LS5, Neuroscience and Disorders of the Nervous System · **Secondary panel:** PE6, Computer Science and Informatics

Abstract

Animal eyes are built for the tasks their owners face, yet it is hard to say which features of a visual circuit exist because of a task: species differ in too much at once. Sexual dimorphism removes this confound. Male hoverflies (Eristalis tenax) chase mates in fast aerial pursuits, and their visual system differs from the female one at every level: a high-acuity dorso-frontal zone, a sexually dimorphic motion-sensitive neuron (HSN) and, as we recently discovered, a male-only third retinal muscle that moves the retina beneath the lenses. HOVER asks which circuit motifs make a visual system task-specific, and whether those motifs are design principles that engineered spiking vision can use. We will (1) build the first connectome-constrained spiking model of a complete gaze loop, from photoreceptors to eye movement, including the electrical synapses we found between motion-sensitive neurons and eye-muscle motor neurons, at three linked levels (biophysical, reduced, trainable) and in both the female and the new male Drosophila connectome; (2) map the model onto male and female hoverflies, test competing circuit hypotheses with optimised controversial stimuli, retinal tracking and muscle recordings, and trace the developmental and genetic origin of the male muscle; (3) train the models on pursuit versus stabilisation under identical wiring and energy budgets, to test whether the male features emerge as task-optimal solutions; (4) transfer the extracted motifs to event-camera spiking vision with task-dependent sensor micro-movements on neuromorphic hardware, benchmarked against deep networks for small-target detection, ego-motion estimation and closed-loop tracking; and (5) validate every model on low-cost spiking robots run by students and citizens. HOVER delivers the first photon-to-eye-movement model of a visual circuit, a mechanistic and normative explanation of a visual sexual dimorphism, and open, tested design principles for efficient active machine vision.

Why a second panel

HOVER asks a question of systems neuroscience: why is the hoverfly visual system sexually dimorphic, and which circuit motifs make it task-specific (LS5_16, LS5_7)? Two of its five objectives, however, are tested in machines. Objective 4 builds event-based spiking vision systems with task-dependent sensor micro-movements and benchmarks them, on public event-camera datasets and on neuromorphic hardware, against deep-network baselines (PE6_11, PE6_14). Objective 5 runs the models on robots at scale. Reviewers with expertise in computer vision and neuromorphic computing are needed to judge the benchmark design, the baselines and the hardware claims of that part of the work.

HOVER: Hoverfly-Optimised Vision through Event-driven Recurrent spiking circuits

PI: Étienne Serbe-Kamp · **Host:** IU Internationale Hochschule (Germany) · **Partners:** MPI for Biological Intelligence, LMU Munich · **Duration:** 60 months · **Panel:** LS5 (secondary PE6)

1. The question

Every visual system is a compromise shaped by the tasks its owner must solve. Flies solve, with about 10^5 neurons and a few milliwatts, problems that machine vision still finds hard: estimating self-motion from optic flow, detecting and chasing a small target in clutter, and keeping the gaze stable. Which features of their circuits exist *because of* a task? Comparing species cannot answer this: species differ in too many ways at once. Evolution has run a cleaner experiment. In the hoverfly, males and females share a genome, a development and a brain plan, yet the male eye is built for one extra task, the high-speed aerial pursuit of mates. **HOVER asks which circuit motifs make a visual system task-specific, and whether those motifs are design principles that engineered spiking vision can use.** It answers in a loop (Fig. A): build a spiking model of the complete fly gaze circuit from the connectome; test its sex-specific predictions in hoverflies; explain the dimorphism by training the model for the male task; transfer the motifs to event-based vision on neuromorphic hardware and flying robots; and validate the models at scale with students and citizens.

2. A natural experiment: one genome, two visual machines

Male hoverflies (*Eristalis tenax*) chase conspecifics in fast aerial pursuits with interception courses [71,72]; females hover, feed and lay eggs. Male specialisations are known at every level. The male eye has a dorso-frontal "bright zone" with larger facets, faster photoreceptors and higher contrast sensitivity [15], the hoverfly version of the "love spot" of blowflies and houseflies, where photoreceptor speed is matched to the demands of chasing [76] and male-specific lobula neurons exist [74,75]. One of the three horizontal-system cells (HSN), a wide-field motion neuron, has its receptive field narrowed to that zone in males, and males carry an additional cell, HSNE [1]. Optic-flow descending neurons are sexually dimorphic at high velocities, although optomotor flight behaviour is not [23]. Hoverflies also have small-target motion detectors of remarkable sensitivity [16,62], and the male pursuit itself has been quantified in the laboratory [21].

Our own recent work adds the missing level: the effector. Flies move their retinas beneath the stationary lenses with small muscles, producing both an optokinetic reflex and spontaneous saccade-like shifts [7]. In a comparative survey of 167 insect species by micro-CT and synchrotron tomography, with behaviour in 59 of them, we found retinal movements only in Diptera, reconstructed their stepwise evolution (spontaneous movements came before the stabilising reflex) and discovered in hoverflies **a male-only third retinal muscle, the M. interocularis (MIO)**, lodged between the two nearly fused male eyes and positioned to shift exactly the bright-zone region of the retina [2]. The male hoverfly therefore has not only a different sensor and a different circuit but a different *active-vision effector*: a complete pursuit channel that females lack. This is a natural knockout experiment with an evolutionary rationale built in: the male features exist for pursuit. In *Drosophila*, the sex-determination genes *fruitless* and *doublesex* are known to rewire visual pathways for male courtship [77,78] and to specify a male-only muscle through its motor neuron [42,43,79]. HOVER uses the hoverfly dimorphism three times: as a test of mechanism (which neurons drive the MIO), as a test of optimality (do the male features emerge in a model trained for pursuit?) and as a source of design principles for machines.

3. State of the art and three gaps

Gap 1. Connectomes have outpaced models, and both miss electrical synapses. Synapse-level wiring diagrams now exist for the female *Drosophila* brain [12,47,48], for brain and nerve cord [49] and, since September 2026, for the complete male central nervous system, with the first synapse-resolution comparison of the sexes [39]. Connectome-constrained networks predict neural activity across the optic lobe from wiring and a task [45], and a whole-brain integrate-and-fire model reproduces sensorimotor processing [46]. All of them, however, (i) use chemical synapses only, because gap junctions are invisible in these electron-microscopy volumes [50], (ii) stop before the motor output and (iii) model one sex of one species in a passive setting. We have shown why this matters for gaze: the motor neurons that move the *Drosophila* retina receive only 3 % of their chemical synapses from visual neurons, yet they are coupled by gap junctions to the HS and VS cells, and this electrical wiring alone predicts their direction tuning, the two-dimensional optokinetic reflex ($R^2 = 0.99$) and an efference copy that flows back into the visual

neurons during spontaneous eye movements [14]. The pathway from photoreceptor to retinal movement is now known cell by cell. To find electrical synapses at connectome scale I built **MESH**, an open pipeline that detects membrane appositions between neuron meshes, validates them in the electron-microscopy volume and serves them for proofreading [41]. It defined the electrical wiring diagram in [14] and, for the 26-neuron LPTC–motor-neuron set, yielded 49 validated contact sites out of 619 candidates.

Gap 2. Active vision is missing from circuit models and from event-based machine vision. Retinal micro-movements refresh the image, defeat photoreceptor adaptation and sharpen acuity [7,51,52]. Event cameras, the neuromorphic sensors that report only brightness changes, have the mirror-image problem: they go blind on static scenes [53], although they now reach automotive products [108]. Engineers have begun to add artificial microsaccades to event cameras with a rotating prism [54], and spiking networks on neuromorphic chips now estimate optic flow and fly drones with a few milliwatts [55,56]. But these movement policies are fixed and task-agnostic, and no model links a sensor movement to the circuit that uses it. The hoverfly MIO is nature's answer to a question engineers have not yet asked: should a pursuit system move its sensor differently from a stabilising one?

Gap 3. Comparing circuit models needs stimuli that tell them apart. Standard gratings cannot discriminate competing circuit hypotheses; stimuli optimised to maximise the disagreement between models can [30]. We built a differentiable bank of Hassenstein–Reichardt and Barlow–Levick motion detectors with ON and OFF pathways, motion opponency [26] and wide-field pooling, and we optimise stimuli by gradient descent to separate models before an animal is used (preliminary result P4, B2).

What machines need from this. Most spiking vision networks are deep networks converted to spikes [64]. They do not use the motifs that make fly vision fast, sparse and robust: linear electrical coupling that pools and transmits signals without synaptic delay [14,92], push-pull opponent coding [26], contrast normalisation that keeps velocity estimates stable in natural scenes [38,96], and active sampling [7]. Whether such motifs transfer is an empirical question. It requires a model grounded in a complete biological loop and a benchmark with matched baselines. HOVER provides both.

4. Objectives and hypotheses

O1. Model. Build a multi-tier, connectome-constrained spiking neural network (SNN) of the fly gaze loop, from photoreceptors to retinal displacement, *including electrical synapses*, in the female [12] and the male [39] *Drosophila* connectome. → **H1 (Sufficiency).** Chemical plus gap-junction connectivity from EM, with gap-junction weights set by apposition area, predicts motor-neuron tuning (± 15°), eye-movement direction and latency; a mathematically reduced SNN with ≥ 10× fewer state variables preserves these predictions. Corollary: the *Drosophila* gaze circuit is monomorphic; the hoverfly's is not.

O2. Test. Map the model onto male and female *Eristalis* and test the dimorphic circuit. → **H2 (Dedicated channel).** In males the dimorphic HSN drives a dedicated motor neuron of the MIO, forming a pursuit-specific active-vision channel; HSN and HSNE also feed the shared stabilisation channel (MOT, MOS) in both sexes. → **H2b (Origin).** The MIO is specified by its motor neuron, as the male-only Muscle of Lawrence of *Drosophila* forms only under *fruitless*-expressing motor neurons [42,43,79].

O3. Explain. → H3 (Task optimality). When connectome-constrained SNNs are trained on the male task (pursuit of small fast targets in clutter) or on the female task (hover stabilisation, feeding approach) under identical wiring priors and energy budgets [80], the male features (narrowed dorso-frontal integration, an extra effector) emerge only under the pursuit objective.

O4. Transfer. → H4 (Transfer). Event-based SNNs built from the extracted motifs, with task-dependent micro-movements of the sensor, outperform parameter- and energy-matched deep-network baselines on small-target detection in clutter and on ego-motion estimation, and achieve closed-loop hovering and pursuit on an indoor quadrotor at a fraction of the energy.

The computer-vision question behind H4. Five predictions are specific enough to fail (H4a–f in B2). (i) A sensor-movement policy tuned to the task improves small-target detection at the same event rate: the male-type policy wins on pursuit, the stabilising policy on ego-motion, and a fixed rotation [54] wins on neither. (ii) Feeding the motor command back into the motion units through the fly's coupling motif, the efference copy of [14], removes self-generated events from the flow estimate better than software compensation. (iii) Linear electrical pooling cuts the latency from first event to motor command without loss of accuracy. (iv) Opponent push-pull coding with contrast normalisation makes velocity estimates robust to contrast and clutter [38]. (v) Under one energy budget, two task-specialised circuits with a shared front end, the dimorphic solution, beat one generalist. Success is measured as accuracy per joule and as latency, against baselines matched in parameters and synaptic operations, under protocols registered before the data exist.

O5. Validate at scale and open everything. Every model runs on low-cost spiking robots in IU computer-vision courses and in a distributed public experiment; students and citizens produce embodied robustness data, proofread electrical synapses and collect hoverfly pursuit statistics.

5. Approach

Five work packages are linked by one modelling ladder and one stimulus engine (Fig. A).

WP1 Connectome → spiking circuit (M1–36). (a) MESH is scaled from 26 neurons to all wide-field motion neurons, retinal and neck motor neurons and optic-flow descending neurons, in the female and the male connectome. A 3D convolutional classifier trained on the expert-validated sites ranks candidates; community proofreading (WP5) confirms them. The male–female comparison in *Drosophila* is the control for H1's corollary. (b) Three model tiers: **Tier 1**, morphologically reduced biophysical neurons with electrical and chemical coupling, fitted by differentiable simulation [66]; **Tier 2**, integrate-and-fire point neurons with gap-junction coupling, derived from Tier 1 by systematic reduction in which Kolmogorov–Arnold networks [93] learn the univariate transfer functions between compartments and return closed-form expressions; **Tier 3**, trainable SNNs whose connectivity is masked by the connectome, trained with surrogate gradients [64,65]. (c) A motor readout maps muscle activation to retinal displacement through the measured pull directions [7,14]. Validation: retinal-motor-neuron imaging, eye tracking and the efference-copy recordings of [14].

WP2 The dimorphic gaze circuit of *Eristalis* (M6–48). No hoverfly connectome exists, so male and female circuits are built by homology from the *Drosophila* model (HSN, HSE, HSS, VS cells, the hoverfly-specific HSNE, and the motor neurons of MOT, MOS and MIO), constrained by our tomography atlas [2], neurobiotin fills and published hoverfly physiology [1,15,23]. This gives **competing male models** (H2 against shared drive, HSNE-driven MIO and VS-driven MIO). The stimulus engine generates the stimuli on which these models disagree most. A HOVER postdoc in the Fenk laboratory (MPI for Biological Intelligence), which holds the only established retinal-tracking and retinal-muscle set-ups, presents them to tethered males and females while tracking the retina through the cornea and recording the three muscles electromyographically. *High risk, high gain:* retinal tracking during free-flight pursuit of artificial targets [21]. With R. Merrill (LMU), a HOVER PhD student traces the developmental origin of the MIO (micro-CT and confocal time series of pupae) and its genetic basis (sex- and stage-resolved transcriptomics, in situ hybridisation chain reaction, the hoverfly genome [44]), testing H2b.

WP3 Theory: why dimorphic? (M12–54). In an embodied simulator with compound-eye optics and retinal actuators, Tier-3 networks are optimised for pursuit and for stabilisation under the same wiring prior and a spike-count energy penalty, both with the option of a third actuator [81,82]. We test whether the male features emerge (H3) and extract the **minimal motif set** by ablation, mean-field analysis of gap-junction-coupled populations and Kolmogorov–Arnold readouts that turn the trained networks into equations.

WP4 Active event-based machine vision (M13–60). The motifs become event-driven SNNs on an event camera mounted on a piezo micro-movement stage that replays fly retinal trajectories, with movement policies that are fixed, fly-derived or learned. Networks are deployed on sensor-integrated and large-scale neuromorphic hardware [67,68,70] through European access programmes. Three benchmarks with pre-registered protocols: small-target detection and tracking in clutter against insect-inspired [61] and deep [94] detectors; ego-motion estimation on public event-camera datasets [58,59] against state-of-the-art artificial and spiking networks [55–57] and the fly-derived velocity-estimation CNN [38]; and closed-loop hovering and target pursuit on an indoor event-camera quadrotor in a motion-capture arena. Metrics: accuracy, latency, synaptic operations and measured energy. All applications are civil, and all code, data and the benchmark leaderboard are open.

WP5 Embodied validation at scale, teaching and open science (M1–60). All Tier-2 and Tier-3 models compile to low-cost two-wheeled camera robots that run spiking neurons natively. (a) Each semester, IU computer-vision students implement and test one HOVER motif; remote students receive a kit by mail. (b) A distributed "robot zoo" experiment: schools, makerspaces and citizens run fly-type, female-type and male-type circuits on standardised pursuit and stabilisation courses and upload logs, yielding thousands of embodied trials under real-world optics and noise. (c) Community proofreading of candidate gap junctions in the MESH viewer, following the EyeWire and FlyWire model [12,85]. (d) "Hoverfly watch": smartphone slow-motion videos of male pursuits in gardens supply the natural target statistics that define the WP3 tasks. WP5 builds on my open citizen-science framework [36] and on a participatory project in which secondary-school pupils became first authors [88].

Feasibility. The pieces exist. MESH runs and has produced validated electrical synapses [14,41]. A Hodgkin–Huxley model of the right-eye gaze circuit with electrical and chemical coupling already reproduces motor-neuron spiking and the direction of optogenetically evoked eye movements. The Tier-1 model, extended by the MIO, already returns a stimulus on which two hypotheses about the MIO's pull direction predict retinal shifts 36° apart (B2, Fig. B1). The tomography atlas of male and female hoverflies is in hand [2], hoverflies can be reared year-round with a published protocol [19], and the hoverfly genome is public [44]. I have shipped TinyML pipelines on classroom robots and teach computer vision to the cohorts that will run WP5. Hypothesis-critical experiments are done by HOVER team members hosted at the partner sites, not by favours.

6. Why now, why me, why here

Why now. Whole-brain connectomes of both sexes, a validated pipeline for electrical synapses, differentiable biophysics, mature surrogate-gradient training, affordable event cameras, sensor-integrated neuromorphic chips and classroom spiking robots have all matured within the last three years. Each piece exists; nobody has closed the loop.

Why me. My path runs through every level HOVER needs, and HOVER is the step that joins them. I spent a decade recording from the fly motion-vision circuit, from the inputs of the elementary motion detector [95] to opponency and real-world velocity coding [26,38,96]. I then moved from one circuit to the whole wiring diagram, built MESH and performed the connectome analysis that revealed the electrical gaze pathway [14,41], and helped trace the evolution of insect eye movements and discover the MIO [2]. In parallel I became an applied computer-vision scientist: segmentation and object recognition in terabyte electron-microscopy volumes, video-based event detection and TinyML classifiers on edge hardware, and, since 2026, Professor of Computer Vision at IU. I study gaze in humans too: my latest first-author work uses mobile eye-tracking and machine learning to read intended behaviour from gaze [89]. And I have helped shape participatory research as a field: first author of the Open Citizen Science framework [36], co-corresponding author of a plant electrophysiology study led by school pupils [88], co-director of a research fellowship spanning more than 20 countries, science-jury member of the German Federal Ministry of Research's IdeenLauf initiative, and invited speaker at citizen-science conferences. HOVER is a new programme that I lead with my own team. It is not a continuation of my postdoctoral projects, nor of the Fenk laboratory's *Drosophila* programme on eye-movement mechanisms, with which it is complementary: HOVER asks *why* a visual circuit differs between the sexes, in a different species, with models and machines as the instruments.

Why here. IU is Germany's largest, largely distance-learning university. A computer-vision professorship there makes WP4 and WP5 core activities, with students, courses and a route to thousands of remote learners with kits at home. The partner laboratories supply what IU cannot: the only hoverfly retinal-muscle physiology set-ups and genomics for a non-model insect.

7. Expected outcomes and impact

(1) The first connectome-constrained spiking model of a complete visuomotor loop, with electrical synapses, from photon to eye movement, in both sexes of *Drosophila*. (2) A mechanistic and normative explanation of a visual sexual dimorphism, including the origin of the only known sexually dimorphic eye muscle. (3) A tested, open catalogue of fly-derived motifs for active event-based vision, with a benchmark that lets the field measure "bio-inspired" claims. (4) A participatory infrastructure that turns connectomics and spiking models into something students and citizens can build, test and contribute to. Beyond flies, HOVER establishes a template for asking *why* a circuit is wired as it is: pit models of different task hypotheses against each other, in animals and in machines.

Ambition and risk. HOVER is high-gain because it joins three things that have not been joined before: a complete connectome-constrained motor loop with electrical synapses, a sexual dimorphism as a normative test, and a machine-vision benchmark as the final arbiter. The riskiest elements, free-flight retinal tracking and the transfer hypothesis H4, are isolated so that their failure does not stop the rest, and a rigorous negative result on H4 would itself be a valuable, open contribution to a field full of untested bio-inspired claims.

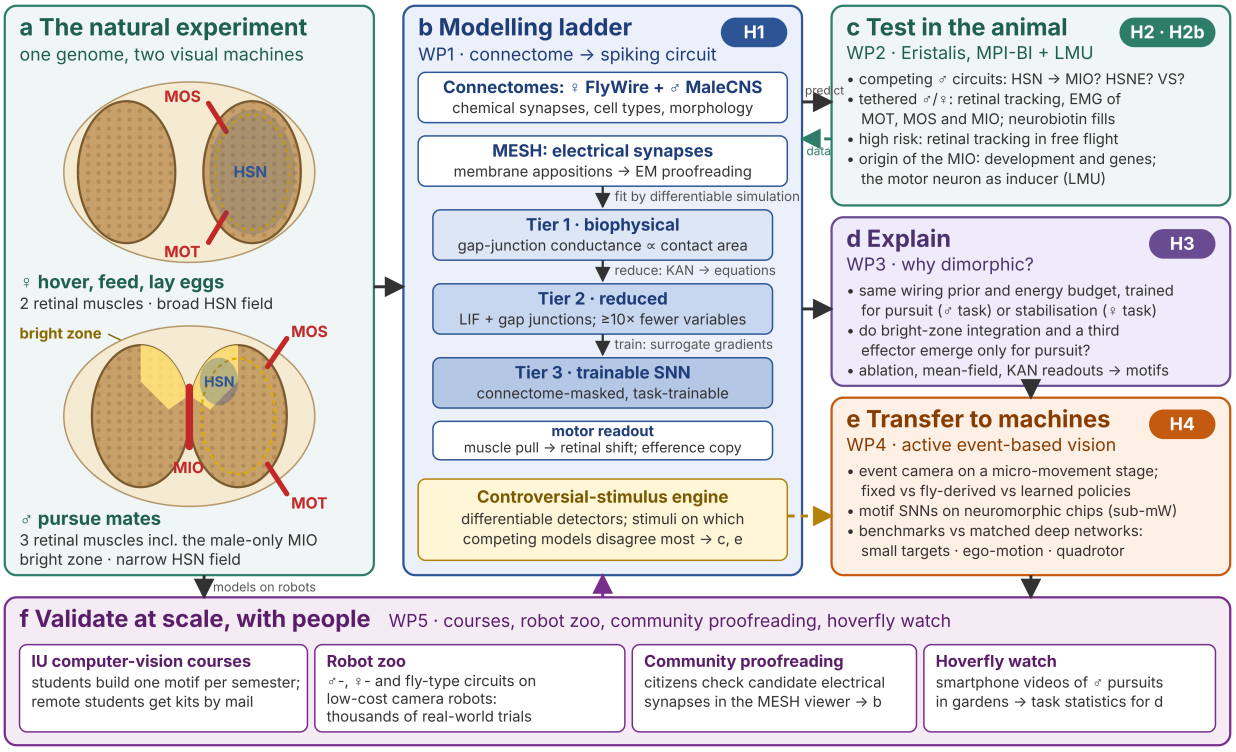


Figure A. *HOVER closes a loop from synapse to society.* Left: the natural experiment, male and female *Eristalis* with the male bright zone, the narrowed HSN receptive field and the male-only third retinal muscle (MIO). Centre: the modelling ladder from the female and male *Drosophila* connectomes and MESH-detected electrical synapses through biophysical (Tier 1), reduced (Tier 2) and trainable (Tier 3) spiking models, with the controversial-stimulus engine. Right: the model is tested in the animal (WP2, H2), explained by task optimisation (WP3, H3) and transferred to active event-based vision on neuromorphic hardware and a quadrotor (WP4, H4). Bottom: students and citizens validate the models on spiking robots, proofread electrical synapses and collect pursuit statistics, which feed back into WP1 and WP3 (WP5).

References (abbreviated; full list in `references.md`)

[1] Nordström *et al.* 2008 Curr Biol · [2] **Kviatko, …, Serbe-Kamp, …, Fenk** 2026 (manuscript) [verify status] · [7] Fenk *et al.* 2022 Nature · [12] Dorkenwald *et al.* 2024 Nature · [14] **Falt, Ammer, Serbe-Kamp *et al.*** 2026 bioRxiv · [15] Straw *et al.* 2006 J Exp Biol · [16] Nordström *et al.* 2006 PLoS Biol · [21] Thyselius *et al.* 2023 J Exp Biol · [23] Nicholas *et al.* 2026 eLife · [26] **Ammer, Serbe-Kamp *et al.*** 2023 Nat Neurosci · [30] Golan *et al.* 2020 PNAS · [36] **Serbe-Kamp *et al.*** 2023 RIO · [38] **Drews *et al.*** 2020 Curr Biol · [39] Berg *et al.* 2026 Cell · [41] **Serbe-Kamp** 2026 MESH software · [42] Lawrence & Johnston 1986 Cell · [43] Usui-Aoki *et al.* 2000 Nat Cell Biol · [44] *Eristalis tenax* genome, Darwin Tree of Life · [45] Lappalainen *et al.* 2024 Nature · [46] Shiu *et al.* 2024 Nature · [47] Schlegel *et al.* 2024 Nature · [48] Matsliah *et al.* 2024 Nature · [49] Bates *et al.* 2026 Nature · [50] Ammer *et al.* 2022 Curr Biol · [51] Juusola *et al.* 2017 eLife · [52] Kemppainen *et al.* 2022 PNAS · [53] Gallego *et al.* 2022 IEEE TPAMI · [54] He *et al.* 2024 Sci Robot · [55] Paredes-Vallés *et al.* 2024 Sci Robot · [56] Hagenaars *et al.* 2021 NeurIPS · [57] Gehrig *et al.* 2021 3DV · [58] Gehrig *et al.* 2021 IEEE RA-L · [59] Zhu *et al.* 2018 IEEE RA-L · [61] Wiederman *et al.* 2008 PLoS ONE · [62] Nordström & O'Carroll 2009 Trends Neurosci · [64] Neftci *et al.* 2019 IEEE SPM · [65] Zenke & Ganguli 2018 Neural Comput · [66] Deistler *et al.* 2025 Nat Methods · [67] Davies *et al.* 2021 Proc IEEE · [68] Mayr *et al.* 2019 arXiv · [70] Yao *et al.* 2024 Nat Commun · [71] Collett & Land 1975 · [72] Collett & Land 1978 · [74] Hausen & Strausfeld 1980 · [75] Gilbert & Strausfeld 1991 · [76] Hornstein *et al.* 2000 · [77] Ribeiro *et al.* 2018 Cell · [78] Hindmarsh Sten *et al.* 2021 Nature · [79] Gailey *et al.* 1991 Development · [80] Niven & Laughlin 2008 J Exp Biol · [81] Yamins & DiCarlo 2016 · [82] Lindsey *et al.* 2019 ICLR · [85] Kim *et al.* 2014 Nature · [88] **Madariaga, …, Serbe-Kamp, Marzullo** 2024 Plant Signal Behav · [89] **Serbe-Kamp *et al.*** 2026 PsyArXiv · [92] Hürkey *et al.* 2023 Nature · [93] Liu *et al.* 2024 (KAN) · [94] Gehrig & Scaramuzza 2023 CVPR · [95] **Serbe *et al.*** 2016 Neuron · [96] **Leonhardt, …, Serbe, …** 2016 Nat Neurosci · [108] Gehrig & Scaramuzza 2024 Nature

Curriculum Vitae and Track Record: Étienne Serbe-Kamp

Personal details

Prof. Dr. Étienne Serbe-Kamp · ORCID 0000-0002-7097-6274 · IU Internationale Hochschule, [campus / address], Germany · [IU e-mail] · github.com/ETigerschuss

Education and key qualifications

- **2016** PhD (Dr. rer. nat.), fast-track programme of the Graduate School of Systemic Neurosciences, LMU Munich, carried out at the Max Planck Institute of Neurobiology. Thesis: *[title]*. Defence: [DD.MM.2016] [verify, the date must match the certificate]. **Supervisor:** Prof. Alexander Borst.
- **2011** BSc Biology, LMU Munich.
- **Key qualifications.** In vivo whole-cell patch clamp, two-photon calcium imaging and optogenetics in the fly visual system; EM connectomics and research-software engineering (MESH); biophysical, reduced and differentiable circuit models; machine learning for stimulus design, segmentation and multimodal time series; edge AI and TinyML on robots; eye tracking and electrophysiology in humans and plants; design and evaluation of open citizen-science projects.

Current and previous positions

- **since 2026** Professor of Computer Vision (W2-equivalent, distance-learning programme Applied Artificial Intelligence), IU Internationale Hochschule, Germany. [start month; verify title]
- **2023–2026** Scientist, Active Sensing group (Dr. Lisa M. Fenk), Max Planck Institute for Biological Intelligence, Martinsried (12.8–50 %): connectome analysis, biophysical modelling, teaching of electrophysiology. [verify end date or ongoing guest status]
- **2026** Lecturer, AI and Bioinformatics (international master's programme, virtual and hybrid), SRH University Heidelberg (April–August).
- **2023–2024** Senior Scientist, Chair of Philosophy of Mind (Prof. Ophelia Deroy), LMU Munich (50 %): mobile eye tracking and machine learning on behavioural and survey data.
- **since 2020** Senior Scientist and Fellowship Co-Director, Backyard Brains Inc., Ann Arbor (USA) and Munich (30–75 %, industry; declared conflict of interest for WP5, see Part B2 and Resources). **2018–2019** R&D Scientist and Fellowship Mentor, Backyard Brains.
- **2016–2022** Postdoctoral researcher, Department Circuits–Computation–Models (Prof. Alexander Borst), Max Planck Institute of Neurobiology, Martinsried (full-time 2016; 25–50 % from 2018 in parallel with the industry position).
- **2011–2016** Doctoral researcher, same department.
- **PhD supervisor:** Alexander Borst. **Postdoctoral mentors:** Alexander Borst, Lisa M. Fenk.

Ten research outputs (most recent first)

Each entry states the significance, my role, and what it demonstrates for HOVER. Joint first authorship is marked *, joint corresponding authorship #.

1. *Falt T, Ammer G,* Serbe-Kamp É, Kroell LM, Friedrich AB, Guerreiro-Mota S, Šimsová E, Fenk LM (2026).** Vectorial efference copy and visuomotor transformation through gap junctions. *bioRxiv*, doi:10.64898/2026.08.11.744090 (posted 17 Aug 2026; in review [verify]). *Significance:* the motor neurons that move the fly retina are coupled to the HS and VS motion neurons by electrical synapses that current connectomes do not capture; the electrical wiring alone predicts motor tuning, the two-dimensional optokinetic reflex and an efference copy. *My role:* I built MESH and performed the connectome-scale contact-site analysis that defined the electrical wiring diagram (Figs. 3 and 4), and collected preliminary neurobiotin fills. *HOVER:* the basis of WP1 and of hypothesis H1.
2. **Serbe-Kamp É, Aliyeva A, Krick CB, Jantus CI, [complete author list exactly as posted] (2026).** Gaze on IPCC climate maps carries behaviour-specific signals for intended low-carbon action. *PsyArXiv*, osf.io/preprints/psyarxiv/64utk (posted Sep 2026; submitted to [journal]) [verify title, author order and DOI]. *Significance:* mobile eye-tracking during free viewing of climate maps, analysed with machine learning, predicts which low-carbon actions people intend. *My role:* first author; conceived the analysis, built the eye-tracking and classification pipeline, wrote the paper [verify]. *HOVER:* independent, first-author work on gaze and behaviour outside the fly field; the data-analysis pipeline carries over to retinal tracking in WP2.

3. **Serbe-Kamp É (2026).** MESH: connectome-scale contact-site detection and EM proofreading for putative gap junctions. Research software, MIT licence. github.com/ETigerschuss/MESH (development) and github.com/ActiveSensing/MESH (release v1.0.0 with a pre-built viewer). *Significance:* the first open pipeline that finds candidate electrical synapses in a whole-brain connectome, validates them in the EM volume and serves them for proofreading; 619 contact sites and 49 validated motor-neuron–LPTC sites for the 26-neuron gaze set, 2,895 EM images, public viewer. *My role:* sole developer. *HOVER:* WP1 and the community-proofreading strand of WP5.
4. **Kviatko P, Renous Y, Falt T, Kroell LM, Guerreiro Mota S, Pujari A, Rimniceanu M, Serbe-Kamp E, Šimsová E, Vollrath F, Wilde F, Hammel JU, Scheel M, Blanke A, Fenk LM (2026).** Spontaneous eye movements preceded visual stability reflexes in the evolution of insects. Manuscript [status: in review / to be posted on bioRxiv before 12 Jan 2027; verify; remove if not public]. *Significance:* comparative anatomy of 167 insect species and behaviour in 59 reconstruct the stepwise evolution of insect eye movements and reveal the male-only third retinal muscle of hoverflies. *My role:* [fill: micro-CT acquisition/segmentation, behavioural assays, analysis]. *HOVER:* the comparative basis of WP2 and hypothesis H2.
5. *Madariaga D, Arro D*, Irarrázaval C*, Soto A, Guerra F, Romero A, Ovalle F, Fedrigolli E, DesRosiers T, Serbe-Kamp É#, Marzullo TC# (2024).** A library of electrophysiological responses in plants: a model of transversal education and open science. *Plant Signal Behav* 19(1):2310977. doi:10.1080/15592324.2024.2310977 [verify the equal-contribution marks]. *Significance:* a library of electrical responses of more than 15 plant species to touch and heat, recorded by secondary-school and undergraduate students on open hardware; the pupils of a school in Santiago de Chile are first authors. *My role:* co-senior and co-corresponding author; designed the study and the recording protocol, supervised the students remotely, analysed the data and wrote the paper with T.M. [verify]. A second paper from this programme is in preparation [fill: status]. *HOVER:* the working model for WP5: participants as researchers and authors, not as an audience.
6. **Ammer G, Serbe-Kamp E, Mauss AS, Richter FG, Fendl S, Borst A (2023).** Multilevel visual motion opponency in *Drosophila*. *Nat Neurosci* 26:1894–1905. *Significance:* motion opponency is implemented at successive stages of the fly motion pathway, which sharpens flow-field selectivity. *My role:* second author; [fill: two-photon imaging and electrophysiology of T4/T5 and LPTCs, analysis, figures; mark joint first authorship if applicable]. *HOVER:* opponency is one of the candidate motifs transferred in WP4.
7. **Serbe-Kamp É, Bemme J, Pollak D, Mayer K (2023).** Open Citizen Science: fostering open knowledge with participation. *Res Ideas Outcomes* 9:e96476. doi:10.3897/rio.9.e96476. *Significance:* defines Open Citizen Science as a methodological basis that joins open-science practice with participation, with two Wikimedia Open Science Fellowship projects as case studies, including ERGo!, my low-cost electroretinogram platform. *My role:* first and corresponding author. *HOVER:* the framework of WP5.
8. **Drews M, Leonhardt A, Pirogova N, Richter F, Schuetzenberger A, Serbe E, Braun L, Borst A (2020).** Dynamic signal compression for robust motion vision in flies. *Curr Biol* 30:209–221 [verify author order]. *Significance:* a fly-like velocity-estimation network trained on natural scenes (`deepfly`) shows that contrast normalisation makes motion vision robust. *My role:* [fill]. *HOVER:* the CNN baseline in WP4 and a motif candidate.
9. **Serbe E, Meier M, Leonhardt A, Borst A (2016).** Comprehensive characterization of the major presynaptic elements to the *Drosophila* OFF motion detector. *Neuron* 89:829–841. *Significance:* the first complete functional characterisation of the inputs to the OFF elementary motion detector by in vivo patch clamp. *My role:* first author; all recordings, analysis and writing. *HOVER:* the physiological ground truth for the Tier-1 front end.
10. **Leonhardt A, Ammer G, Meier M, Serbe E, Bahl A, Borst A (2016).** Asymmetry of *Drosophila* ON and OFF motion detectors enhances real-world velocity estimation. *Nat Neurosci* 19:706–715. *Significance:* a biological design choice, the ON/OFF asymmetry, improves velocity estimation in natural scenes, which is exactly the logic of HOVER's H4. *My role:* [fill: electrophysiology].

Further outputs: co-author of the directional tuning map of elementary motion detectors (Maisak *et al.* 2013 *Nature*), of contrast vision (Bahl *et al.* 2015 *Neuron*), feedforward inhibition (Mauss *et al.* 2014 *J Neurosci*) and complementary mechanisms of direction selectivity (Haag *et al.* 2016 *eLife*); more than 20 peer-reviewed papers in total.

Selected examples of significant peer recognition

- **2026** Appointment as Professor of Computer Vision, IU Internationale Hochschule, after a competitive call (Applied Artificial Intelligence programme).
- **2026** Keynote lecture, GSN Student Conference, Graduate School of Systemic Neurosciences, Munich [verify].
- **2025** Invited speaker, international conference *Citizen Science: Bridging Theory and Practice*, Vilnius University (talk: *Citizen Science, in and out of the Lab*); panellist, European Citizen Science Association webinar *AI and Citizen Science*.
- **2024** LMUexcellence Innovation Fund, LMU Munich [role and amount; verify].
- **2022** Member of the science jury of *IdeenLauf*, the citizen-participation initiative of the German Federal Ministry of Education and Research (BMBF) in the Science Year 2022.
- **2022–2023** Principal investigator of two EU Horizon Europe proposals scored 14/15 and 14.5/15 [state whether funded; verify].
- **2020** Chen Innovator Grant, Tianqiao and Chrissy Chen Institute, Caltech [role; verify].
- **2024–** Scientific lead for modelling and machine learning on the NIH SBIR phase II project *SpikerBots* (brain-based neurorobots for schools; PI G. Gage, Backyard Brains) [verify].
- **[year]** Fellow, Open Science Fellows Program (*Fellow-Programm Freies Wissen*), Wikimedia Deutschland and Stifterverband [verify year]. **[year]** Young Scientist Award [fill body].
- Media: scientist in season 2 of the ZDF/KiKA series *Princess of Science* (the series received the Grimme-Preis), ProSieben *Galileo*, Radio Television of Serbia; more than 50 public science events on four continents.

Supervision, teaching and mentoring

- Co-director (since 2020; mentor since 2018) of the Backyard Brains Fellowship, an international research programme: more than 30 fellows from more than 20 countries, supervised almost entirely online; fellowship projects have led to peer-reviewed papers with fellows as authors [n; verify].
- Supervision of [n] BSc and MSc theses and co-supervision of [n] doctoral researchers [fill].
- Teaching: computer-vision modules at IU (since 2026); applied AI, computer vision and statistics in the international master's programme at SRH Heidelberg (2026); lecturer at the Munich Neurocamp (2019, 2025); summer schools and workshops since 2011, including more than 50 school and public workshops on four continents.

Invited talks and lectures (selection)

2026 keynote, GSN Student Conference, Munich [verify] · 2025 international citizen-science conference, Vilnius University · 2025 ECSA webinar on AI and citizen science · 2024 VISTA Science Academy, Klosterneuburg · 2023 *Conduction Unveiled*, international online lecture · 2022 SENSE festival, BIOTOPIA Munich · 2021 Wikimedia Winter School · 2020 innOsci *Innovations in Open Science*, Berlin · 2019 Hirnkastl workshop, Cambridge · 2017 Brain Awareness Week, Montevideo.

Funding obtained or co-obtained

LMUexcellence Innovation Fund (2024) [role, amount] · Chen Innovator Grant, Caltech (2020) [role] · Open Science Fellowship, Wikimedia Deutschland [year, amount] · scientific lead for modelling on the NIH SBIR phase II project *SpikerBots* (PI G. Gage) [verify] · PI of two EU Horizon Europe proposals (2022, 2023) scored 14/15 and 14.5/15 [funded? verify].

Career path and life events

My career combines academic research (Max Planck Society, LMU), industry (Backyard Brains, a neuroscience-education company, where I co-direct an international research fellowship) and science communication. From 2018 I held the Max Planck position part-time alongside the industry role, which is why the postdoctoral period spans six calendar years. This combination is the capacity that WP4 and WP5 require: I have shipped edge-AI products, mentored more than 30 scientists and engineers in more than 20 countries almost entirely online, and turned public participation into peer-reviewed research. [Add parental leave or other life events if applicable.]

Noteworthy contributions to the research community

- **Open research tools:** MESH (connectome contact detection and EM proofreading, with pre-built releases that need no installation); ERGo! (open electroretinogram platform); open models of the fly

motion pathway. Co-development of the NIH-funded Computational SpikerBox and SpikerBot, low-cost platforms on which pupils and students run spiking neurons and computer vision.

- **Participatory research:** co-founder of the Hirnkastl Science Collective; author of the Open Citizen Science framework; a published study with school pupils as first authors; Open Science ambassador (BMBF IdeenLauf). Lecturer at the Munich Neurocamp (2019, 2025) and in international summer schools since 2017. [Add reviewing and editorial activities.]

Research independence and the capacity to deliver HOVER

My work spans the levels HOVER connects: cellular physiology of fly motion vision (first-author *Neuron* 2016; *Nature Neuroscience* 2016, 2023), connectomics and research software (MESH, the electrical gaze pathway), evolution (the origin of insect eye movements and the male-only third retinal muscle), differentiable modelling and machine learning (stimulus optimisation, segmentation of terabyte EM volumes, TinyML on robots), human gaze (first-author eye-tracking work at LMU) and participatory science (first and senior-author publications, a national jury, international invited talks). The independent lines are visible: MESH is my own tool, the citizen-science programme is my own, the eye-tracking study is my own, and the gap-junction search strategy that made the electrical gaze pathway findable was my design. HOVER is a new programme defined and led from my group at IU. It moves from describing circuits to asking *why* circuits differ between the sexes, in a new species, with connectome-constrained spiking models, neuromorphic machines and public experiments as the instruments. It is complementary to, and agreed with, the Fenk laboratory's *Drosophila* programme on eye-movement mechanisms. The partner groups contribute set-ups and expertise; the team members at the partner sites are selected, funded and directed by me.